

Mengyuan (Millie) Wu

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RESEARCH INTERESTS

My research focuses on designing, engineering, and evaluating **AI agentic systems** that interact with human and their **multimodal context** to support **personalization, productivity, and well-being**. I am especially interested in building efficient, robust, and adaptive systems that can operate persistently and be meaningfully **deployed and adopted in real-world settings**.

AI Agents, Multimodal Sensing, Human-Computer Interaction

EDUCATION

- 2026 – Current **Columbia University**, New York, NY
Ph.D. in Computer Science, Advised by: **Prof. Xuhai “Orson” Xu**
- 2023-2025 **Columbia University**, New York, NY
M.S. in Computer Science | *GPA: 4.02*
- 2017-2019 **Columbia University**, New York, NY
B.S. in Operations Research (Major) & Psychology (Minor) | *GPA: 3.94*

PUBLICATION

- 2026 **Wu, Mengyuan**, Jiang, Z., Fan, Y., Feng, R., Dharmavaram, S., Fallon, S., Polowitz, M., Islam, B., Benson, L., Tung, I., Creswell, J. D., Xu, X. *MindfulAgents: Personalizing Mindfulness Meditation via an Expert-Aligned Multi-Agent System*. **ACM CHI 2026 (Best Paper Honorable Mention Award, top 5%)**
- Li, Z., He, X., **Wu, Mengyuan**, Tong, Z., Wei, H., Yang, B., Feiner, S., Sajda, P. *SwEYEpinch: Exploring Intuitive, Efficient Text Entry for Extended Reality via Eye and Hand Tracking*. **ACM CHI 2026**
- Jiang, Z., **Wu, Mengyuan**, Zou, R., Xu, S., Macmaunus, E., Liao, S., Zhang, P., Wang, D., David, J.L., El-Bassel, N., Mamykina, L., Levin, F.R., Sultan, R., Xu, X. *PAWS: Empowering Everyday Cannabis Use Disorder Support through a Personalized AI Digital Pet on Smartwatches*. **ACM CHI 2026 Workshop (Best Poster Award)**
- Li, Z., He, X., Chen, H., Zuo, J., Bai, S., Yang, B., **Wu, Mengyuan**, Ledner, J., Cheng, Y., Yamauchi, A., Turakhia, D.G., Feiner, S., Sajda, P. *Augmenting Human*

Performance with an XR Agent Learning from Online Behavior and BCI Evidence.
ACM UIST 2026

Submitted

Fan, Y., **Wu, Mengyuan**, Chen, Q., Liu, X., Yang, Y., Zhang, Z., Nepal, S., Nie, J., Jiang, X., Xu, X. *TimeSRL: Generalizable Time-Series Behavioral Modeling via Semantic RL-Tuned LLMs — A Case Study in Mental Health*. Submitted to **IMWUT 2026** (Under Major Revision).

Sun, Y., et al. *Agents' Last Exam*. Submitted to **NeurIPS 2026 Evaluations and Datasets Track. (Co-author)**

Zhang, Z. *, Selvendran, V. *, **Wu, Mengyuan** *Sharpen Before You Adapt: Data-Free Entry-State Sharpening for Test-Time Reinforcement Learning*. Submitted to **NeurIPS 2026 Workshop**

Jiang, Z. *, **Wu, Mengyuan** *, Zou, R., Xu, S., Xu, X. *Deco: Making Physical Companions Pervasive through a Dual-Embodiment Framework*. Submitted to **ACM CHI 2027**

Wu, Mengyuan, Guan, X. *, Nam, Y. *, Chung, R., Wang, K., Wang, Z., Xu, Z., Babbitt, E., Bloom, P., Funkhouser, C., Mishra, S., Xu, X. *"Live Your Day, Twice": A Multi-Modal Agentic Mental Rehearsal Framework for Closing the Intention-Action Gap*. Submitted to **ACM CHI 2027**

Chung, R., Li, M. *, **Wu, Mengyuan** *, Pollack, A., Pratt, W., Mamykina, L., Xu, X. *ExTENDS: Empowering Domain Experts to Build Personalized AI Proxies for Asynchronous Patient Care*. Submitted to **ACM CHI 2027**

PATENTS

Mengyuan Wu, Mathew W. Polowitz, John D. Creswell, Shawn M. Fallon. Method for Generating a Meditation Protocol. U.S. Provisional Patent Application No. 63/991,687, filed Feb. 26, 2026. Assignee: Equa Health, Inc.

SELECTED PRESS

2026 *"Mental Health: Unpacking the AI Revolution in Therapeutic Support."* **Sci PaperMill**

AWARDS & HONORS

2026 **ACM CHI 2026 Honorable Mention Award** (top 5%)

2024 **2nd Place of Intel AI Track Hackathon** (Hoohacks, University of Virginia)

1st Place of NeuReality AR/VR Competition (Columbia University)

2017-2019 **Tau Beta Pi Engineering Honors** (Columbia University, top 1/8th)

RESEARCH EXPERIENCE

2024 - Current **SEA Lab, Columbia University** New York, NY
Research Assistant, Mentor: Prof. Xuhai “Orson” Xu

2024 **LIINC and CGUI, Columbia University** New York, NY
Research Assistant, Mentor: Prof. Steve Feiner and Prof. Paul Sadjia

INDUSTRY EXPERIENCE

2025 - Current **Equa Health (CMU Funded Start-up)**, Full-Stack Software Engineer (Part-time)

2021 - 2023 **Evercore Partners**, Vice President of Investment Research

POSTER & PRESENTATIONS

2025 **AI Agents for Work, DAP Lab**, presented poster
AI at VP&S Workshop: Foundation Models Across Scales: From Cells to Health Systems, presented poster

TEACHING

Fall 2026 **COMS W4995: The Future of Personal Health Assistant** (Teaching Assistant)

Fall 2024 **COMS W4995: Applied Machine Learning** (Course Assistant)

Spring 2019 **IEOR E3609: Advanced Optimization** (Course Assistant)

MENTORING

2026 **Xin Guan** (Tsinghua University B.S.)

Irene Nam (Columbia University B.S., M.S.)

Annie Wang (Columbia University B.S.)

2025 **Akito Yamauchi** (Columbia University B.S.)

Zeyi Tong (Columbia University B.S.)

PAPER REVIEW

2026 **IEEE Internet Computing**

ACM CHI Conference on Human Factors in Computing Systems (2027)

ACM Symposium on User Interface Software and Technology (UIST)
(2026)

**Proceedings of the ACM on Interactive, Mobile, Wearable and
Ubiquitous Technologies (IMWUT)**

2025 **ACM CHI Conference on Human Factors in Computing Systems** (2026)

ACM Transactions on Computing for Healthcare

SKILLS

Research

Data Analysis: Python, PyTorch, MATLAB, R

Methods: Qualitative, Quantitative and Mixed Methods

User Research: Longitudinal Study, Formative Study, User
Interaction Design, User Surveys, Thematic Analysis,
Interviews, Participatory Design

Prototyping

Software: React, Typescript, Javascript, HTML, FastAPI,
Langchain, Google Cloud, Firebase, Git

Hardware: Wearable Sensor Integration, EEG, Arduino,
Micro-controllers, Circuit Prototyping

Design: Figma, Adobe Creative Tool Kits

OTHER INTEREST

Interdisciplinary Reading (Philosophy, Literature, Psychology) | **Road**

Cycling | **Vocals** (Country, Jazz)